

$$\boxed{V_1 = 13.2 \text{ kV} \angle 0^\circ \text{ V}}$$

	T_1	T_2
Connection	Yy0	Yy0
Voltages	13.2 kV / 132 kV	138 kV / 69 kV
S_n	5 MVA	10 MVA
$Z_{eq} \leftarrow V_{sc}$	10%	8%
P_{sc}	50 kW	80 kW
P_o	15 kW	20 kW
$Y_T \leftarrow I_o$	2%	1.5%

Solution: Parameters in "absolute" values.

• Generator

$$\bar{Z}_b = \frac{(13.2 \text{ kV})^2}{5 \text{ M}} = 34.848 \Omega$$

$$X_s = 0.2 \times \bar{Z}_b = 6.97 \Omega$$

$$\Rightarrow \boxed{\bar{X}_s = 6.97j \Omega}$$

• T_1

- Nominal current w/r to HV.

$$I_n = \frac{5 \text{ M}}{\sqrt{3} \times 132 \text{ k}} = 21.869 \text{ A}$$

- Shortcircuit:

$$10\% \cdot V_n = Z_{eq} I_n \rightarrow Z_{eq} = \frac{10\% \cdot V_n}{\sqrt{3} I_n}$$

$$Z_{eq} = 348.48 \Omega$$

$$50 \text{ k} = 3 \cdot R_{eq} \cdot I_n^2$$

$$\Rightarrow R_{eq} = 34.848 \Omega$$

$$\Rightarrow X_{eq} = 346.733 \Omega$$

$$\boxed{\bar{Z}_{eq} = 34.848 + 346.733j \Omega}$$

- Open circuit

$$2\% \cdot I_n = \frac{V_n}{\sqrt{3}} Y_T \Rightarrow Y_T = 5.739 \cdot 10^{-6} \text{ S}$$

$$15 \text{ k} = 3 \left(\frac{132 \text{ k}}{\sqrt{3}} \right)^2 G_{fe} \Rightarrow G_{fe} = 0.8609 \cdot 10^{-6} \text{ S}$$

$$\Rightarrow B_m = 5.674 \cdot 10^{-6} \text{ S}$$

$$\Rightarrow \boxed{\bar{Y}_T = 0.8609 \cdot 10^{-6} - 5.674 \cdot 10^{-6}j \text{ S}}$$

• T_2

- Nominal current w/r to HV.

$$I_n = \frac{10 \text{ M}}{\sqrt{3} \times 138 \text{ k}} = 41.837 \text{ A}$$

- Shortcircuit:

$$8\% \cdot V_n = Z_{eq} I_n \Rightarrow Z_{eq} = 152.352 \Omega$$

$$80 \text{ k} = 3 R_{eq} I_n^2 \Rightarrow R_{eq} = 15.235 \Omega$$

$$\Rightarrow X_{eq} = 151.588 \Omega \Rightarrow \boxed{\bar{Z}_{eq} = 15.235 + 151.588j \Omega}$$

- Open circuit:

$$1.5\% \cdot I_n = \frac{V_n}{\sqrt{3}} Y_T \Rightarrow Y_T = 7.876 \cdot 10^{-6} \text{ S}$$

$$20 \text{ k} = 3 \left(\frac{138 \text{ k}}{\sqrt{3}} \right)^2 G_{fe} \Rightarrow G_{fe} = 1.05 \cdot 10^{-6} \text{ S}$$

$$\Rightarrow B_m = 7.806 \cdot 10^{-6} \text{ S}$$

$$\Rightarrow \boxed{\bar{Y}_T = 1.05 \cdot 10^{-6} - 7.806 \cdot 10^{-6}j \text{ S}}$$

• Line

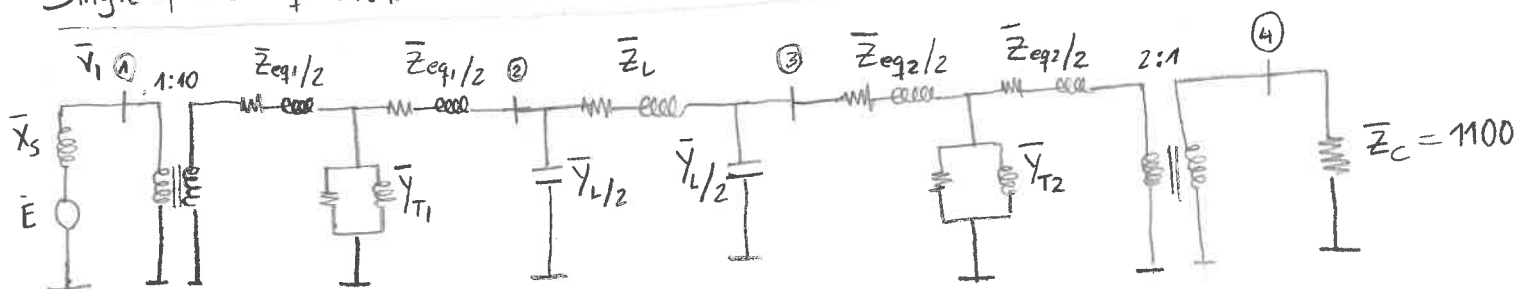
$$\bar{Z}_L = 10 + j100 \, \Omega$$

$$\bar{Y}_L = 3 \times 10^{-4} \, S$$

• Load

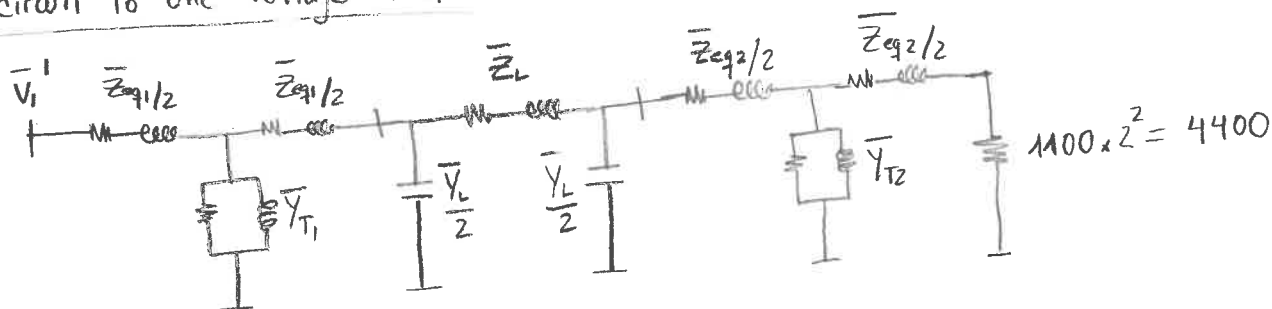
$$\bar{Z}_C = 1100 \, \Omega$$

Single-phase equivalent circuit of the three-phase system



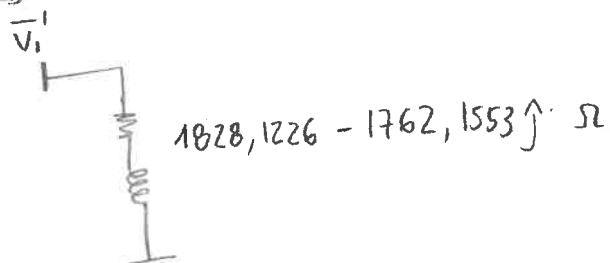
$$\bar{V}_1 = \frac{13,2 \text{ kV}}{\sqrt{3}}$$

Moving circuit to one voltage level



$$\bar{V}_1 = \frac{132 \text{ kV}}{\sqrt{3}}$$

Reducing circuit



$$\Rightarrow \bar{I}_1' = 30,014 \angle 43,9474^\circ$$

$$\bar{S} = 3 \bar{V} \bar{I}^*$$

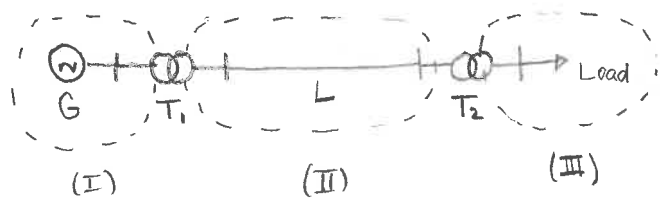
$$P_G = 4,9406 \text{ MW}$$

$$Q_G = -4,7623 \text{ MVAR}$$

$$\Rightarrow \bar{V}_4' = 83,692 \angle -9,03^\circ \text{ V}$$

$$P_C = 4,776 \text{ MW}$$

Solution: Per-unit system $S_b = 10 \text{ MVA}$



	T_1	T_2
$P_{sc} \text{ (P.U.)} \rightarrow r_{eq}$	0,01	0,008
$P_0 \text{ (P.U.)} \rightarrow g_{fe}$	0,003	0,002

Zone I

$$\rightarrow V_b = 13,2 \text{ kV}$$

$$Z_b = \frac{(13,2 \text{ k})^2}{10 \text{ M}} = 17,424 \Omega$$

$$I_b = \frac{10 \text{ M}}{\sqrt{3} \times 13,2 \text{ k}} = 437,387 \text{ A}$$

Zone II

$$\rightarrow V_b = 13,2 \text{ k} \left(\frac{132}{13,2} \right) = 132 \text{ kV}$$

$$Z_b = \frac{(132 \text{ k})^2}{10 \text{ M}} = 1742,4 \Omega$$

$$I_b = \frac{10 \text{ M}}{\sqrt{3} \times 132 \text{ k}} = 43,739 \text{ A}$$

Zone III

$$\rightarrow V_b = 132 \text{ k} \left(\frac{69}{138} \right) = 66 \text{ kV}$$

$$Z_b = \frac{(66 \text{ k})^2}{10 \text{ M}} = 435,6 \Omega$$

$$I_b = \frac{10 \text{ M}}{\sqrt{3} \times 66 \text{ k}} = 87,477 \text{ A}$$

Convert values to the new p.u. system

• Generator:

$$X_s = 0,2 \left[\frac{(13,2 \text{ k})^2}{5 \text{ M}} \right] \left[\frac{10 \text{ M}}{(13,2 \text{ k})^2} \right] = 0,4$$

$$\Rightarrow \bar{X}_s = 0,4 \uparrow$$

• T_1

$$Z_{eq} = 0,1 \left[\frac{(13,2 \text{ k})^2}{5 \text{ M}} \right] \left[\frac{10 \text{ M}}{(13,2 \text{ k})^2} \right] = 0,2 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} X_{eq} = 0,199$$

$$r_{eq} = 0,01 \left[\begin{array}{c} \downarrow \\ \text{(Same)} \end{array} \right] = 0,02$$

$$Y_T = 0,2 \left[\frac{5 \text{ M}}{(13,2 \text{ k})^2} \right] \left[\frac{(13,2 \text{ k})^2}{10 \text{ M}} \right] = 0,01$$

$$g_{fe} = 0,03 \left[\begin{array}{c} \downarrow \\ \text{(Same)} \end{array} \right] = 0,0015$$

$$\Rightarrow \bar{Z}_{eq} = 0,02 + 0,199 \uparrow$$

$$\Rightarrow \bar{Y}_T = 0,0015 + 9,887 \times 10^{-3} \uparrow$$

• T_2 :

$$Z_{eq} = 0,08 \left[\frac{(138 \text{ k})^2}{10 \text{ M}} \right] \left[\frac{10 \text{ M}}{(132 \text{ k})^2} \right] = 0,0874$$

$$r_{eq} = 0,008 \left[\begin{array}{c} \text{"} \end{array} \right] = 0,0087$$

$$\Rightarrow X_{eq} = 0,0870$$

$$\Rightarrow \bar{Z}_{eq} = 0,0087 + 0,0870 \uparrow$$

$$Y_T = 0,015 \left[\frac{10 \text{ M}}{(138 \text{ k})^2} \right] \left[\frac{(132 \text{ k})^2}{10 \text{ M}} \right] = 13,724 \times 10^{-3}$$

$$g_{fe} = 0,002 \left[\begin{array}{c} \text{"} \end{array} \right] = 1,8299 \times 10^{-3}$$

$$\Rightarrow b_m = 13,601 \times 10^{-3}$$

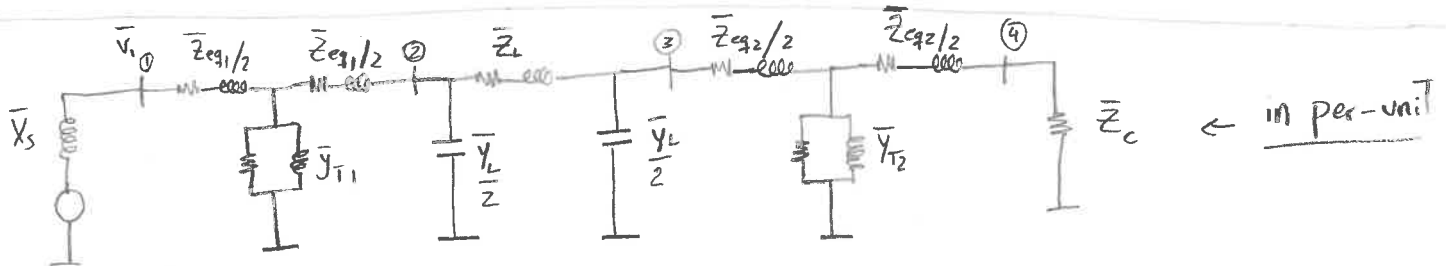
$$\Rightarrow \bar{Y}_T = 1,8299 \times 10^{-3} - 13,601 \times 10^{-3} \uparrow$$

• Line : $\bar{Z}_L = \frac{10 + j100}{1742,4} = 5,739 \times 10^{-3} + j57,392 \times 10^{-3}$

$\bar{Y}_L = \frac{3 \times 10^{-4}}{1/1742,4} = 0,5227j$

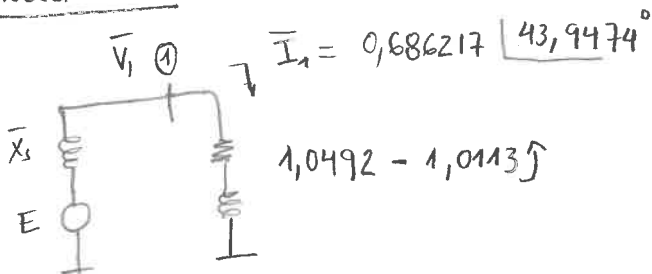
• Load :

$\bar{Z}_C = \frac{1100}{435,6} = 2,525$



$\bar{V}_1 = 1 \angle 0^\circ$

Reduced circuit



$\bar{S} = \bar{V} \bar{I}^* = 0,494061 - j0,476233j$

$P_G = 4,9406 \text{ MW}$

$Q_G = -4,7623 \text{ MVAR}$

$\bar{V}_4 = 1,098167 \angle -9,0313^\circ$

$\bar{I}_4 = 0,43487 \angle -9,0313^\circ$

$\bar{S} = 0,4776 \text{ (P.U.)}$

$P_C = 4,776 \text{ MW}$

• Per-unit system benefits:

- Different voltage levels (due to transformers) disappear.
- "Single-phase" equations for a three-phase system.